

# Section 7

## Emotion & Learning

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COGS 17

12/05/25



**NEXT WEEK (MON Dec 8)**

**3:00-4:20 Midterm 3 (125 points)**  
**AND 4:30-5:50 Final Exam (100 points)**

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# Emotion

# Theories of Emotion



# Theories of Emotion

## James-Lange Theory (1880s)

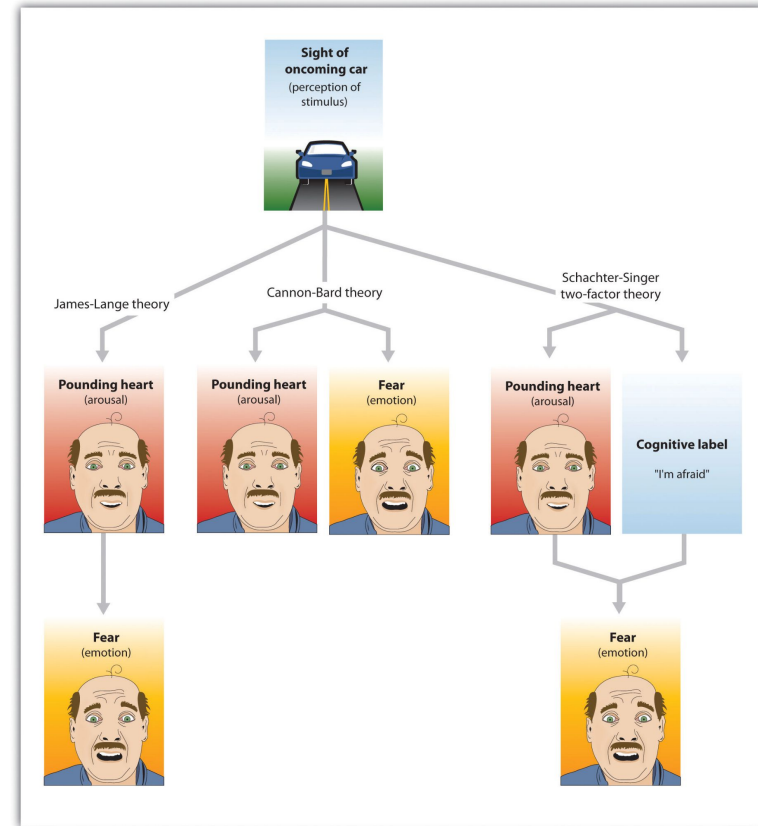
- Emotion = After-the-fact-label assigned to physiological arousal
- Subjective “feelings” are an interpretation we make of our body’s reaction to stimuli ( 🚗 → ❤️ → 😱 )

## Cannon-Bard Theory (1930s)

- Once threat is perceived, visceral & subjective experience of emotion is simultaneous ( 🚗 → ❤️ + 😱 )
- via Perceptual input via Thalamus to Cortex and activation of the ANS (for somatic responses)

## Schachter-Singer Theory (1980s)

- Contemporary model based on neuroscience
- Emotion = Interaction between cognitive appraisal and autonomic/limbic activity ( 🚗 → ❤️ + 🧠 → 😱 )
- Physiology determines how strong emotion is, but identifying emotion depends on a cognitive appraisal of situation



# Theories of Emotion

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Q. The **Cannon-Bard** theory proposed that...

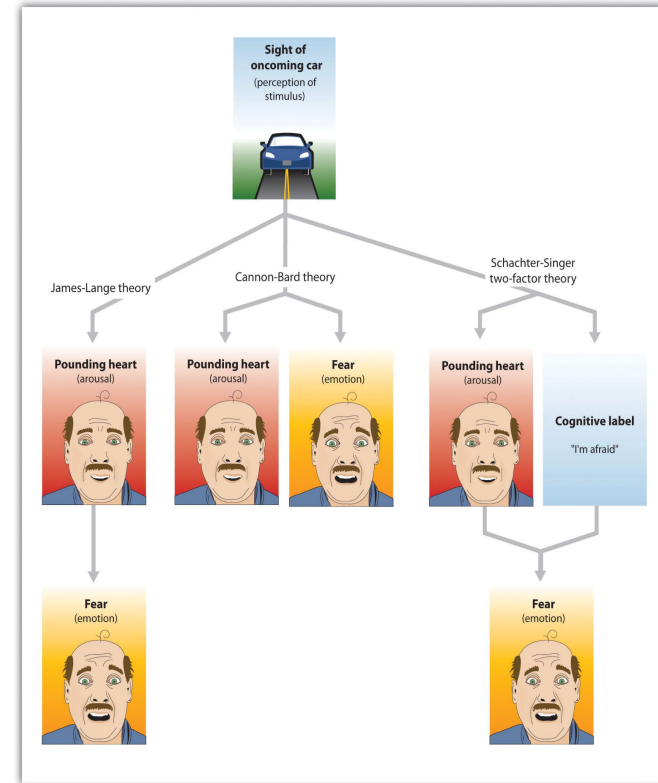
- a. the brain arouses the body and triggers emotions simultaneously.
- b. the physiological arousal occurs first, triggering emotions.
- c. emotions trigger a physiological reaction in the body.
- d. physiological reactions and thinking work together to trigger emotions.

# Theories of Emotion

Q. The **Cannon-Bard** theory proposed that...

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+ Q. Can you find descriptions for **James-Lange & Schacter-Singer theories**?



# Emotional Behavior

- Judgments of Emotional Stimuli
  - Subjects injected with arousing drug (amphetamine) have an exaggerated emotional response to stimuli
- **Facial Feedback**
  - Holding pen in teeth to put face muscles in smile-like configuration → judge comics as funnier
  - Subjects with specific muscle changes (e.g., “Raise brows”) without using emotional terms → reported “feeling” emotions and physiological changes as well (e.g., fear: heart rate up, skin temperature down)

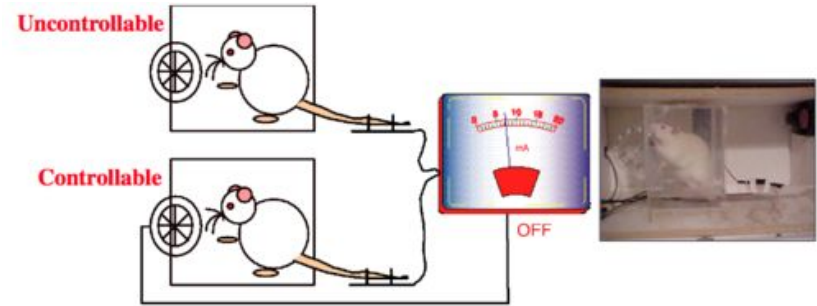




# Emotional Behavior

## Learned Helplessness

- Condition 1: A rat with a running wheel learns that it can stop the shock stimuli by running in wheel
- Condition 2: The rat given the same shock but without wheel option → becomes depressed ('There's nothing I can do...' or such appraisal of a situation) and developed ulcers from the stress of the shocks
- If the prefrontal cortex is removed, the rat in Condition 2 will not become depressed or develop ulcers → appraisal requires higher level cortical activity



Two groups of rats receive tailshock from the same source, The controllable-shock group can terminate the shock by turning a wheel that is mounted in the front of the chamber. The uncontrollable-shock group receives exactly the same amount of shock, but cannot terminate the shock.

# Expressing Emotions

## Universals

: 6 basic emotions with corresponding facial expressions

### Neonatal imitation

- basic emotional expressions mimicked by newborn
- We are pre-wired to practice manipulating the muscles of emotional expression based on behavior of others

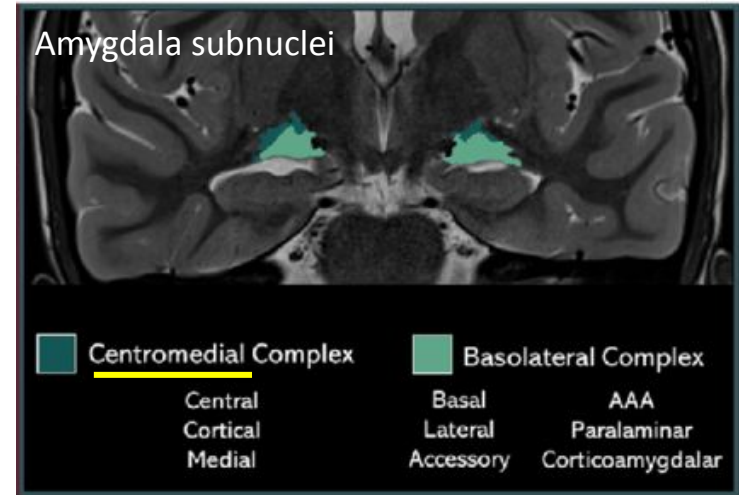
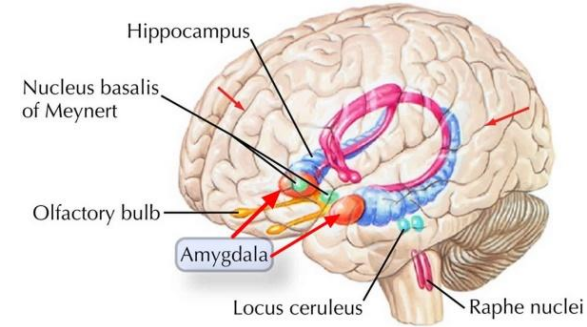
... with culture-specific “display rules” for when and whom you may show what kinds and the extent of emotions

→ may depend in part on learned associations



# The Limbic System: Amygdala

- ‘Emotional Center’ of the brain: critical in producing & interpreting emotions
  - Next to the anterior horn of the hippocampus, directly connected to basal ganglia
  - Multiple nuclei with various functions and patterns of connections
1. **Corticomedial area**
    - Direct stimulation leads to anger and rage - most primal emotion that is critical to survival
    - Rabies virus likely affects this area

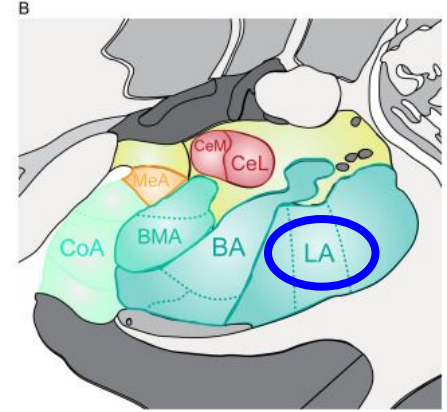


# The Limbic System: Amygdala

## 2. Lateral Nuclei

: Startle reflex (e.g., jumping and hunching shoulder, pushing arms from the floor) is influenced by amygdala connections

- 1) from pain fibers and Visual/Auditory input: to trigger startle reflex, and to detect and learn negative associations
- 2) to Central gray (= periaqueductal gray) area of midbrain
  - part of tegmentum for motor control, esp. neck muscles
  - e.g., clench these muscles to help protect fragile cervical neurons near surface
- 3) to hypothalamus: influences Autonomic NS response (e.g., increase blood pressure, heart rate)



# The Limbic System: Amygdala

## 3. Central and Basolateral Nuclei

- for “conditioned fear” via integrating sensory info (e.g., vision + pain)
- Unlearned startle reflex (e.g. loud noise !!) can be modified via learning
  - Light 💡 + paired with a noxious stimulus (e.g., shock ⚡), light alone does not elicit Startle Reflex
  - but... once association is learned, shock enhances Startle Reflex (i.e. jump higher to noise if light on)
  - alternatively, when associated with pleasant stimulus, it will decrease Startle Reflex (e.g., 💡 + soothing warmth ☀️ → not jump as high to noise when the light is on)

>> Emotionally-laden stimuli (if it's not too intense) is remembered better via connections to/from hippocampus

TRAINING: LIGHT and SHOCK PAIRED



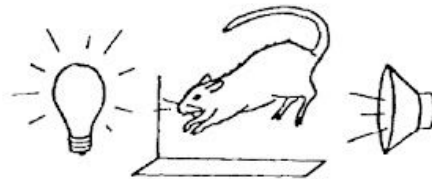
TESTING:

NOISE-ALONE TRIALS



NORMAL STARTLE (in dark)

LIGHT-NOISE TRIALS



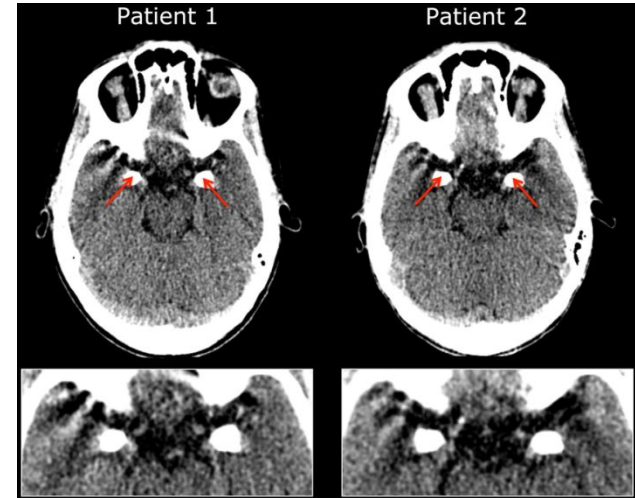
POTENTIATED STARTLE (in light)

# The Limbic System: Amygdala

- Shared Emotion: Amygdala is not just active when feel/express but also when observe emotion in others
- **Urbach-Wiethe Disease** involves the calcification of the amygdala

## Symptoms

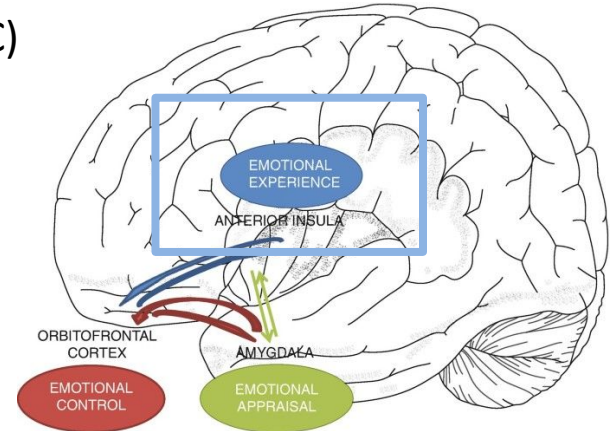
- 1) Patients exhibit a “flattening of affect” with less emotional expression, generally does not experience strong likes/dislikes
- 2) Impaired ability to recognize facial expressions, esp. fear and untrustworthiness
- 3) Recognizes individuals, gender, but have trouble naming emotions, judging if face is “approachable” or not



bilateral calcification damage to the amygdala in Urbach–Wiethe patients

# Anterior Insula

- on the path from amygdala to the orbitofrontal cortex (OFC)
- connected to hindbrain cranial nerves for control of facial muscles
- includes primary Gustatory Cortex
- vital for **emotional expression in social contexts**

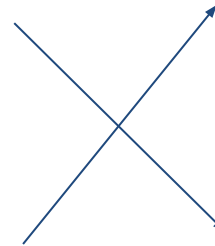


## “Emotional Facial Paresis”

- Damage to the left anterior insula
- Can produce a full smile on command but doesn't spontaneously smile in a funny social situation (esp. right side of face)

## vs. “Volitional Facial Paresis”

- Damage to the right motor cortex
- Cannot produce a smile on command but can spontaneously smile (esp. left side of face)



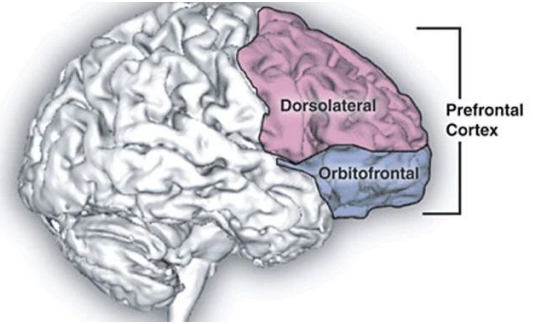
On command

Spontaneous



# Prefrontal Cortex

- Area anterior to the premotor cortex
- Particularly important for regulating social and emotional behavior (executive functions)
- Damage will lead to loss of inhibition, socially inappropriate behavior, sudden aggression, etc
- **Phineas Gage**
  - Sep 13th, 1848 (25 yo), was when accident happened where Steel rod damaged most of PFC
  - Cognitive abilities remain intact but radical change in personality



*"Popular reports of Gage often depict him as a hardworking, pleasant man before the accident. Post-accident, these reports describe him as a changed man, suggesting that the injury had transformed him into a surly, aggressive heavy drinker who was unable to hold down a job, many of his acquaintances explained that after the injury, he was "no longer Gage."*



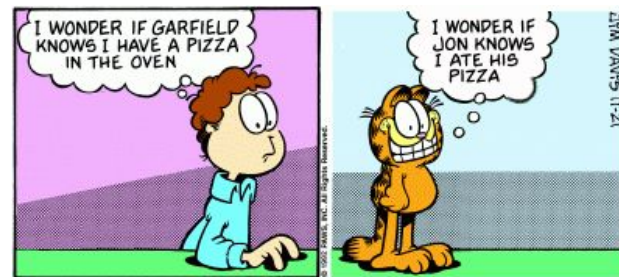
# Phineas Gage



# Functions of Prefrontal Cortex

- **Theory of Mind**

- Ability to attribute mental states (knowledge, feelings, etc) to self and others
- Inappropriate reciprocal connections between the OFC-amygdala have been implicated in Autism



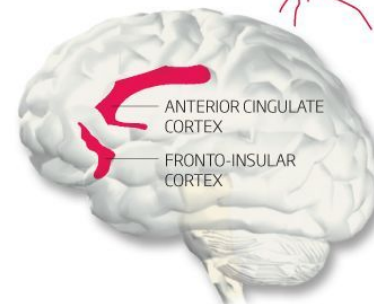
- **Von Economo Cells, ‘Spindle’ Cells**

- Specialized pyramidal neurons with long fibers but few branches
- communicate b/w distant brain areas, with little intervening influence
- e.g., connect Anterior Insula (social emotion) with Anterior Cingulate Cortex (social risk, cost/benefit analyses)
- Found only in large brained animals (humans, elephants, whales 🧑🏫 🐘 🐳)

## VON ECONOMO NEURONS

Allows the high-speed connections necessary for rapid emotional and intuitive judgements

These cells are found in just two small areas of the brain



# Gambling Task

- Compare groups: Normal Control, Amygdala-Lesioned, Prefrontal-Lesioned
- "Pick cards from one of 4 piles to learn about each pile's payoff/penalty"

## • Normal Control

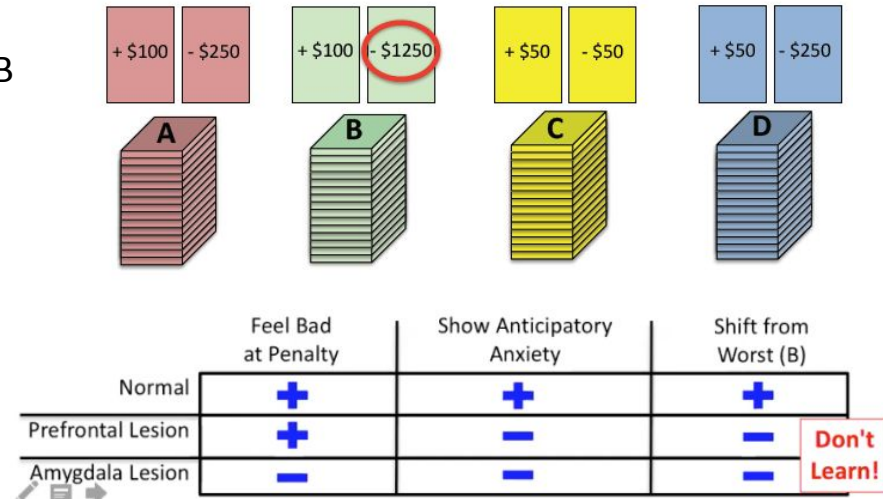
- Shows anticipatory anxiety for picking from Pile B
- Feels bad at penalty
- shift to better, even before can explain decision

## • Prefrontal Lesioned

- X anticipatory anxiety or shift
- Some negative emotion towards penalties
- Does not learn to avoid Pile B

## • Amygdala Lesioned

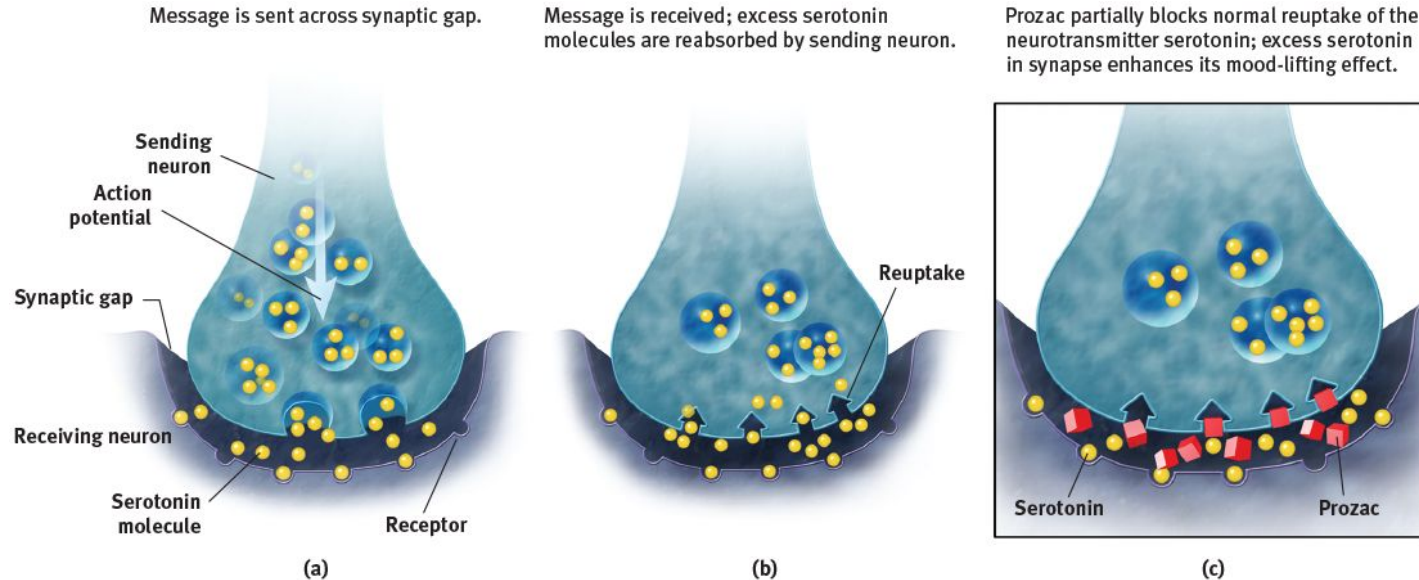
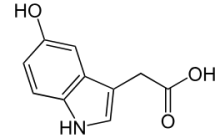
- X anticipatory anxiety
- X negative emotion towards penalties
- Does not learn to avoid Pile B



# Neurotransmitters and Emotion

## 1. Serotonin (5HT) in Amygdala

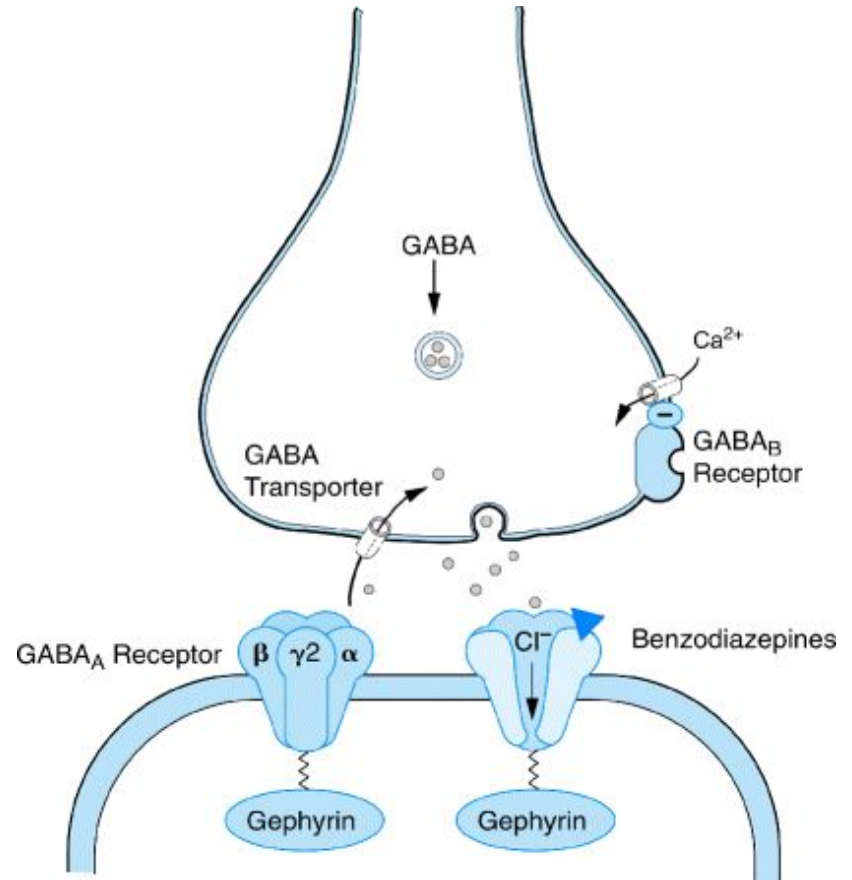
- Reuptake of Serotonin into presynaptic cell, creates a metabolic byproduct 5-HIAA
- Low levels of 5-HIAA in bloodstream: linked w/ impulsivity and depression
- **Antidepressants** like Prozac **blocks reuptake of 5HT**, keeping **serotonin active in synapses longer**
- But, long-term use will ultimately make you produce less serotonin



# Neurotransmitters and Emotion

## 2. GABA

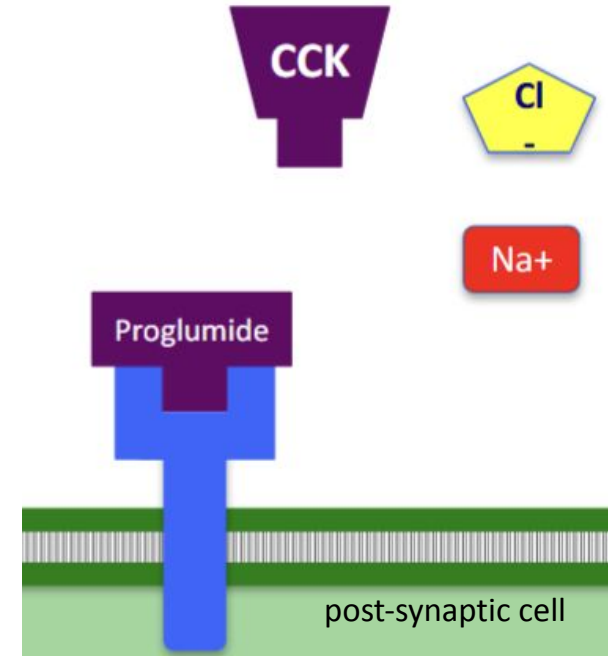
- Inhibitory effect on post-synaptic cell by admitting  $\text{Cl}^-$  ions (hyperpolarized cell, more negative)
- in amygdala, this helps suppresses startle reflex and reduce anxiety
- Anti-anxiety drugs like Benzodiazepines (e.g., Valium) act as GABA agonists
  - BZD binds to GABA receptors, increasing inhibitory effect of GABA (more  $\text{Cl}^-$  coming into the cell)
  - enables GABA to bind more easily and longer



# Neurotransmitters and Emotion

## 3. Cholecystokinin (CCK)

- Stimulates post-synaptic cells by opening  $\text{Na}^+$  gates
- CCK in amygdala: involved in learned enhancement of startle reflex
- CCK in hypothalamus: suppresses hunger when blood sugar rises
- Some diet pills are CCK agonists that mimic CCK effects
  - Side effects involve increases levels of anxiety
- CCK-antagonists (e.g., Proglumide for treating Ulcers) can block receptor sites w/o opening  $\text{Na}^+$  gates in the hypothalamus. This has a calming effect in the amygdala but promotes overeating





# The Science of Emotion in <Inside Out>

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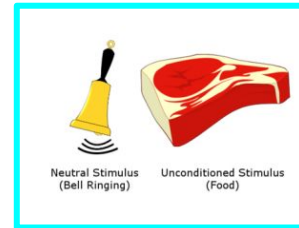
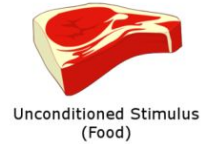
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# Learning & Memory



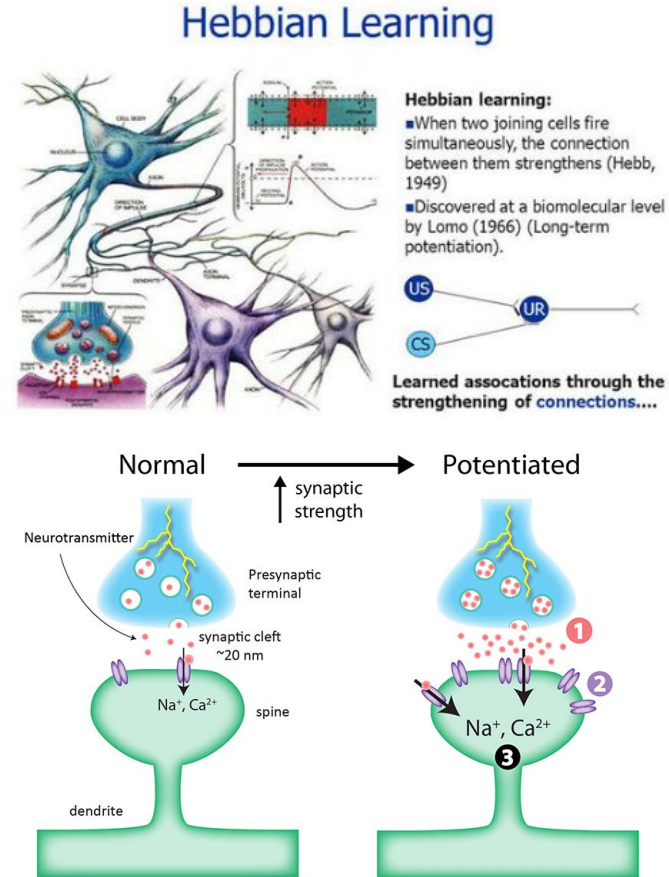
# Learning

- Definition: Development of a permanent change in behavior based on experience
- **Law of Effect:** Any stimulus/action/context associated with positive reinforcement will tend to repeat
- **Classical Conditioning (1897)**
  - Learning happens when the two stimuli are linked together
  - Pavlov's Dog
  - Bell & Food must **co-occur** for the conditioning to succeed
- **Temporal Contiguity**
  - Both types of conditioning require co-occurrence to become associated in the learner's mind
  - Proposed that co-occurrence leads to neural co-activity of the stimulated circuits, thus leading to learning



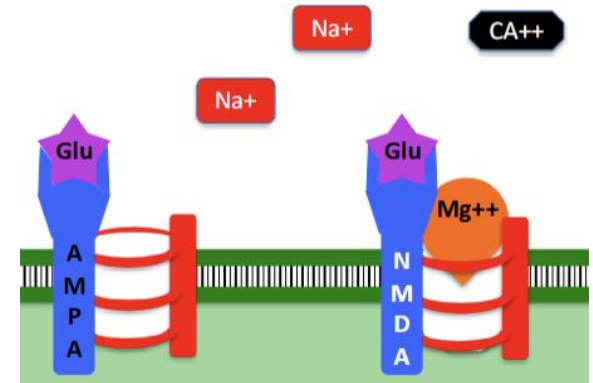
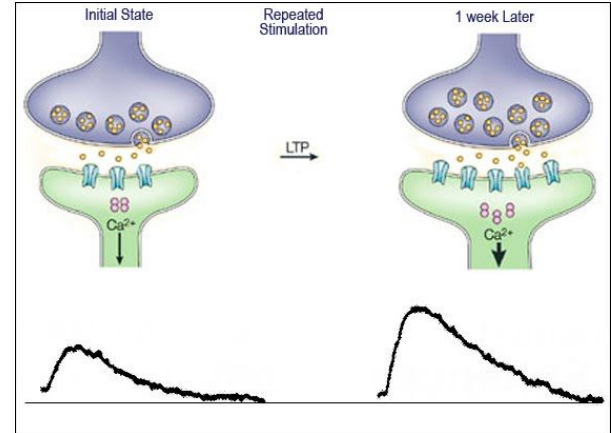
# Hebbian Learning

- Proposed by Hebb as the fundamental neural process involved in learning
- Neurons that fire together, wire together**
  - Repeated co-activated neural circuits, involved in learning and retrieval of associations, are reinforced and increases the likelihood of circuits co-firing. **“Hebbian Synapse”**
  - by structural/metabolic changes in NT availability, release, and reception
- Long-Term Potentiation (LTP)**
  - Key mechanism underlying learning
  - Over time, LTP can lead to (semi-) permanent structural and connectivity changes among neurons
  - Ultimately, increases the likelihood of activity along repeated circuits

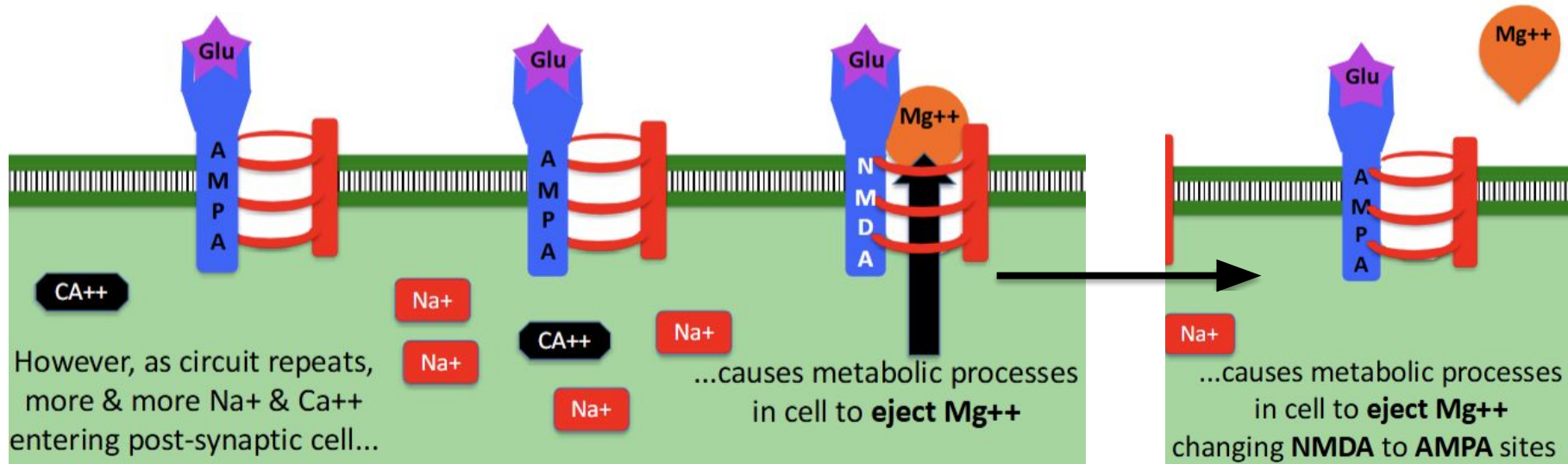


# Long-Term Potentiation (LTP)

- Typically changes occur on the post-synaptic neuron
  - Changes to # of receptor sites
  - Dendritization (building new dendritic branches), etc.
- The best-studied mechanisms of LTP use **Glutamate** as the primary NT
- 2 types of Receptor Sites for Glu
  - AMPA Receptors**
    - Glu attaches → ion gates open to admit  $\text{Na}^+$  and  $\text{Ca}^{++}$
    - Pumps  $\text{Mg}^{++}$  out of the cell/receptor sites
  - NMDA Receptors**
    - In contrast to AMPA receptors, when Glu attaches to NMDA sites  $\text{Mg}^{++}$  blocks the NMDA channel
- But... as circuit repeats**, more  $\text{Na}^+$  and  $\text{Ca}^{++}$  enter the post-synaptic cell → causes metabolic processes in cell to eject  $\text{Mg}^{++}$
- Result:** 1) NMDA receptors are converted into AMPA, 2) build new dendritic branches



# Long-Term Potentiation (LTP)



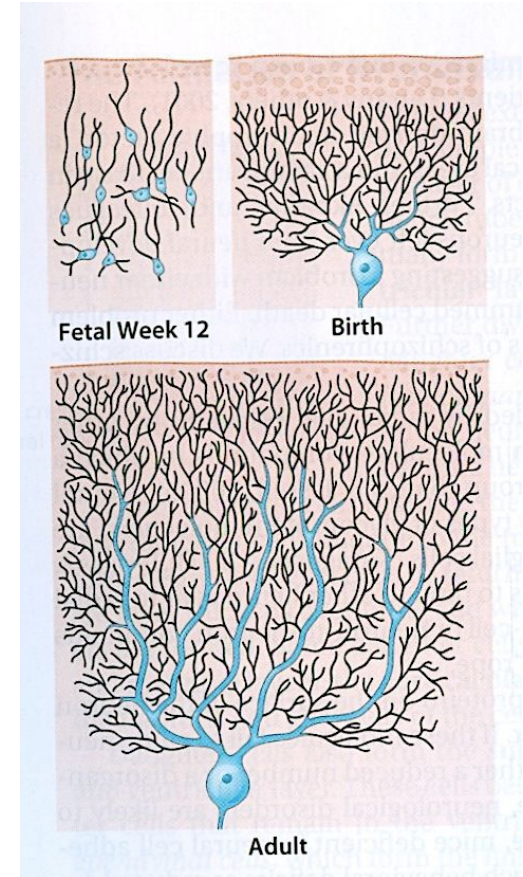
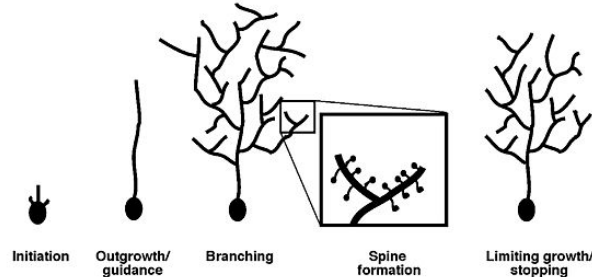
# Long-Term Potentiation (LTP)





# Dendritization

- Significant experiences (e.g., learning new languages, how to play guitar etc) over time changes how neuron looks like
- How? It introduces physical changes in cells: increasing **# of branches** which increase the **# of receptor sites**
- Repetition is the key: Continued activity along circuit → growth of new dendritic branches
- In some memory circuits, **retrograde messengers** (e.g., Nitrous oxide) are released by post-synaptic cells, throughout life, to create a positive feedback loop and prolong release of NTs by pre-synaptic cells
  - RECALL: exceptional case which does not usually happen after birth



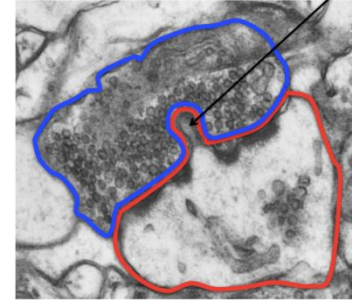
# Perforation (Splitting of Synapses)

- Occurs where the terminal button meets the axons of the next cell
- Post-synaptic (red) branch will grow protuberance** pokes into the pre-synaptic (blue) terminal (membrane of the pre-synaptic terminal is stretched but not broken)
- Results of Perforation
  - 1) duplicates the metabolic mechanisms in both sides
  - 2) promotes the **division of the pre-synaptic terminal into two terminal buttons**
- The dendrites then dismantles the protuberance and divides into two dendritic spines that each receive NT from one of the new terminal buttons

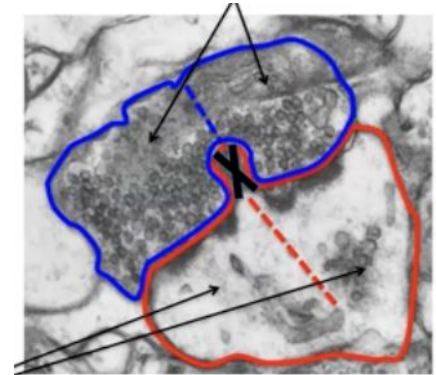
## Long-Term Potentiation: **Perforation**

Post-Synaptic cell builds a temporary protuberance that deforms Pre-Synaptic terminal

Pre-Synaptic Terminal  
("perforated" –  
membrane stretched,  
not broken)



Post-Synaptic dendritic spine  
(with protuberance  
that "perforates"  
pre-synaptic terminal)



# Other Factors

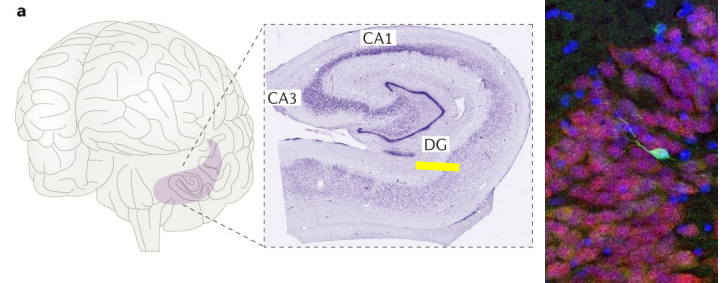
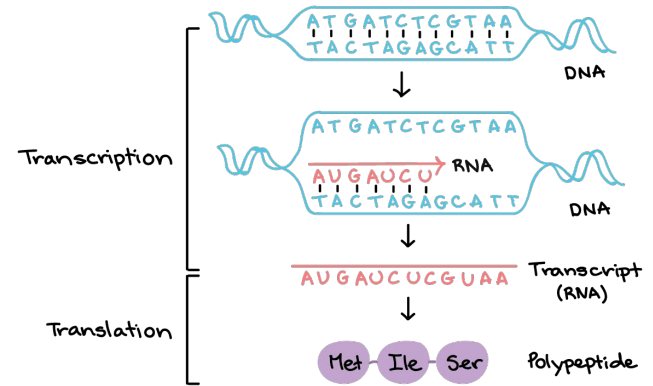
...that modify function based on experience

- **Genes**

- Have impact through gene transcription (makes copies of segment of DNA (=RNA) to code for protein production and transport it to ribosome to build protein)
- These proteins can change the # of available NTs, the size and distribution of vesicles, and metabolic processes in the cells

- **Neurogenesis**

- The growth of new neurons in the adult brain
- Very rare and is only known to occur in sparse regions
  - Subgranular Zone (SGZ) in the dentate gyrus (DG) of the hippocampus
  - Subventricular Zone (SVZ) of the lateral ventricles



Left: Young neurons (green) in the human hippocampus and more mature neurons (red). Image credit: Sorrells et al.



# Memory

- Often divided into 3 classes with critical brain regions for each type (though very loosely defined and not always localized to one region)

**Very Important!** Memories are not stored in these regions, but are vital for their retrieval and consolidation

## 1. Hippocampus

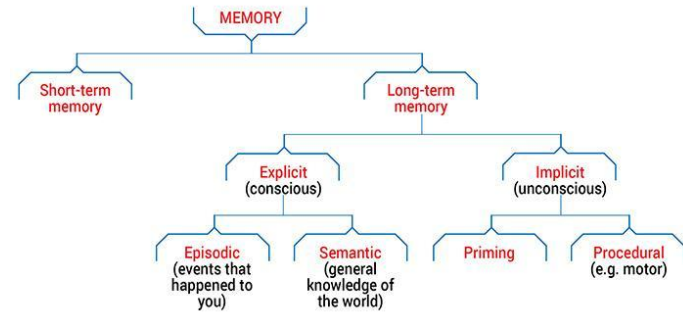
- Spatial Memory (e.g., “where is my phone, where is York Hall”)
- Develops place cells over the course of spatial learning → cognitive maps of where the animal is in space and where locations are relative to the animal
- Damage can impair navigation, map reading, recall of locations

## 2. Cerebellum and Basal Ganglia

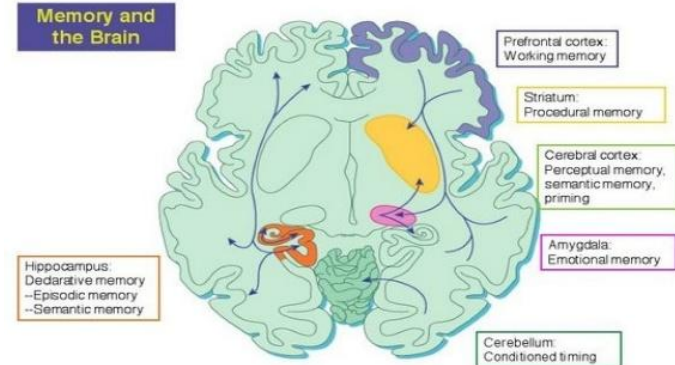
- Procedural memory: Sequences of behavior that you’ve learned how to do (e.g., riding bicycle)

## 3. Hippocampus, Medio-dorsal Thalamus, Prefrontal Cortex

- Declarative memory: facts or personal episodes (e.g., rules for tennis game, what I did during the last weekend)

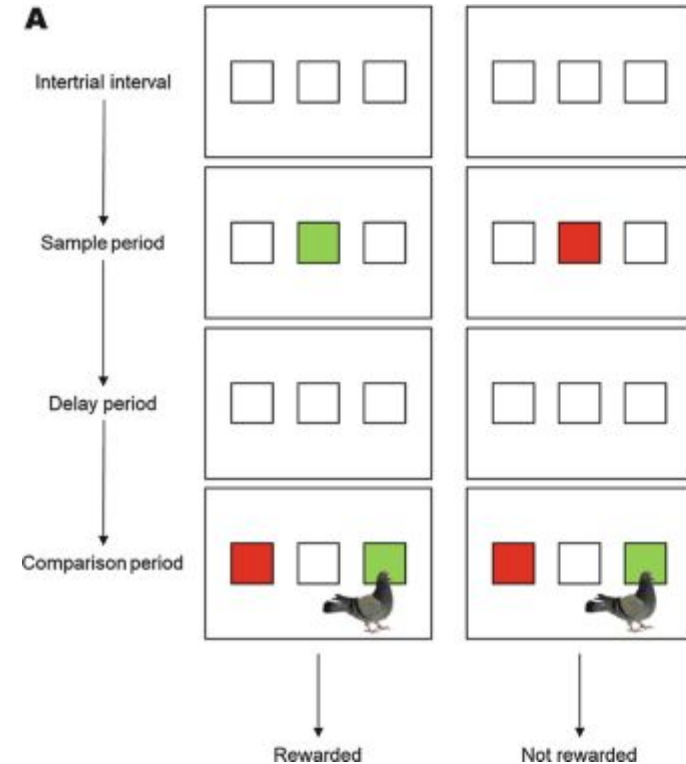


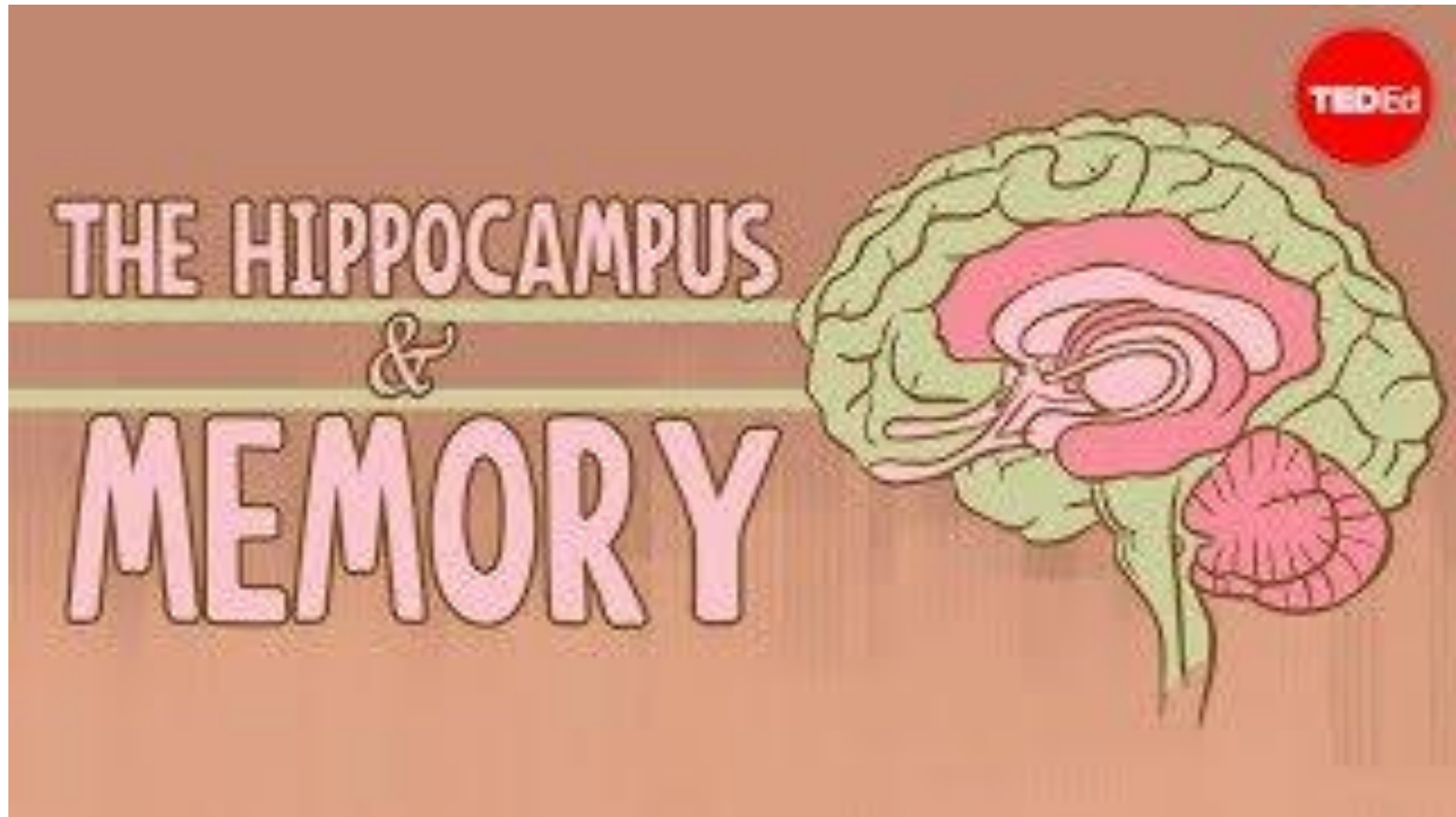
Different regions of brain involved in specific memory



# Declarative Memory

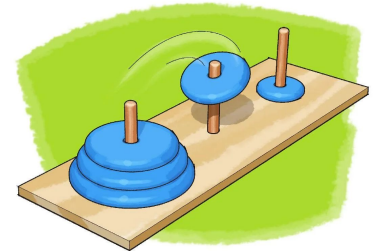
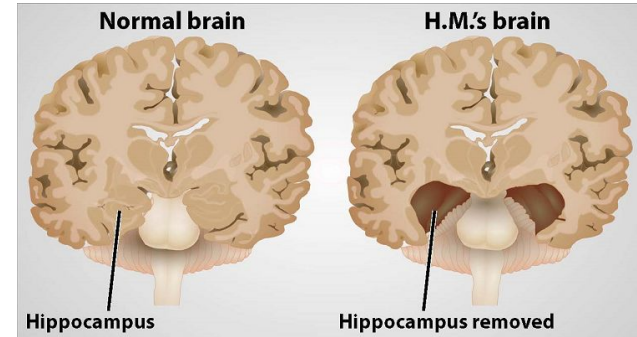
- Memory of “facts”, abstract “rules”, and “episodic memory”
  - Involves the Prefrontal Cortex, Hippocampus, and Medio-dorsal Thalamus (Medio-dorsal Nuclei, MDN)
  - **Match-to-Sample Task:** a declarative memory task that doesn't require language
    - A cue is given and give a reward when animal matches the choice to the sample in a task phase
    - Performance is impaired if the hippocampus is lesioned after the learning the rule
- 💡 Hippocampus plays a vital role in forming and retrieving memories
- If connections from the MDN to the PFC are damaged (as in B1 deficiency from chronic alcoholism) Korsakoff's Syndrome develops
    - Suffers from anterograde amnesia (inability to form new memories after specific accident or injury) and confabulation (can't remember → make up a story based on current cues)





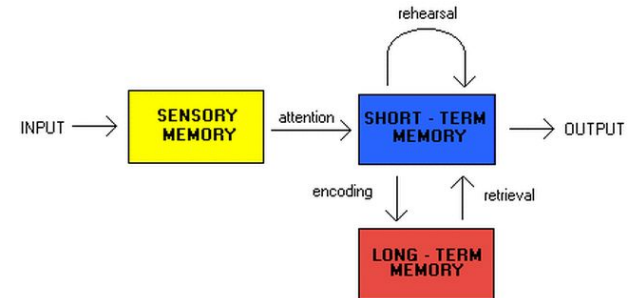
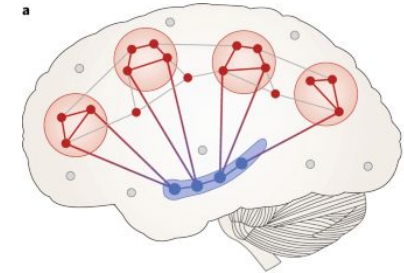
# Patient H.M.

- Bilateral Hippocampus, amygdala, and partial regions of the temporal cortex were removed to treat H.M.'s epileptic seizures
- Results led to reduction of epilepsy but other symptoms arose
- H.M.'s IQ and personality was intact, but suffered from severe anterograde amnesia (e.g., after meeting a new people, no memory of having met them)
  - Was unable to form new episodic memories after his hippocampal resection, but Procedural Memory and Working Memory was largely intact
  - With repetitive training, H.M. learned new skills (like solving the “Tower of Hanoi” puzzle) but was not able to recall having learned them



# Long-Term Memory

- Declarative memory is **not** stored in the Hippocampus, Thalamus, or Prefrontal Circuits
- However these circuits are required for the consolidation and retrieval of information that are stored elsewhere in the cortex
  - Area of ongoing research, it is generally believed that memories are distributed to various circuits throughout the cortex
- Fusiform Gyrus of the Inferior Temporal Cortex is active in recognizing faces (Fusiform Facial Area, FFA)
- Dorsal Temporal Cortex including Wernicke's Area is used for recognizing words and voices
- Medial Parietal (e.g., Precuneus) has some spatial memory for how you engage with the world from an egocentric POV



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So close to the end of the quarter now!

